

UNIT - II

TEACHING SKILLS

2.1 Micro Teaching

A Teacher make use of a number of methods and techniques to make the teaching effective, successful and interesting. The technique includes motivating, explaining, questioning, writing on the black board, using teaching aids and so on. The teacher could also make use of non verbal behaviours such as smiling, nodding and gesturing. These group of activities are called skills. So teaching is a various combination of skills.

Microteaching introduces to the teacher trainee to a wide range of teaching skills and allows the teacher trainee to practice each skill one at a time until he or she becomes proficient in the skill.

2.1.1 Microteaching - Definition

The word Microteaching was formed by A.W.Dwight Allen during 1963 defined microteaching "as a scaled down teaching encounter in class size and class time".

Allen Eve defined microteaching "as a system of controlled practice, that makes it possible to concentrate on specific teaching behaviour and to practice teaching under controlled conditions."

Microteaching is reduced teaching. Here new skills are polished - M.C. Knight 1971.

Microteaching is a teacher training technique. Here a teacher trainee teachers a short lesson of about five minutes duration to a small group of students practicing one teaching skill at a time.

2.1.2 Need and importance of Microteaching

In teaching learning process, we can't use single skill. At the same time if a teacher wants to be good in teaching, he should be given training in all teaching skills one by one. For this micro teaching is more useful.

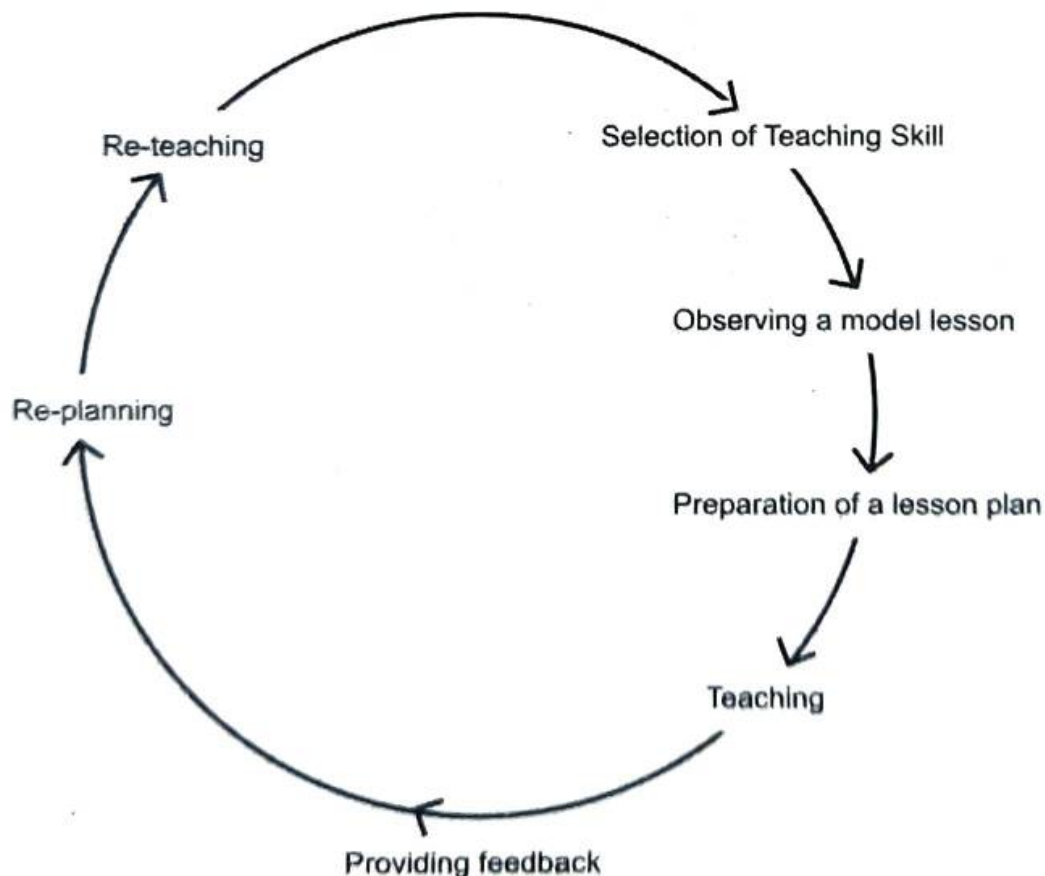
Here each skill is looked liked through microscope. Instead of "What to teach", how to teach is considered. The beauty of this method is, teacher trainee get unproblematic artificial atmosphere instead of natural class room atmosphere. This method is very helpful to improve the teaching skills and to remove the difficulties in teaching.

2.1.3 Characteristics of microteaching

- Microteaching is a teacher teaching technique and not a teaching method.
- It is a real teaching but if focuses on developing teaching skill and competencies.
- In microteaching the teacher trainee practices one specific teaching skill at a time, till he/she attains mastery over the skill.
- Class size is reduced to 5 - 6 students.
- Duration of the class is 5 - 6 minutes
- Microteaching is a cyclic process, the predecided model is plan, teach, feedback, re-plan, re-feedback. It is repeated till the trainee achieves the expected level of mastery.
- Microteaching is practiced in terms of definable, observable, measurable and controllable teaching skills.

2.1.4 Microteaching Cycle

Microteaching is a method that enables a teacher - trainee to practice one teaching skill till he or she attains mastery over the skill. The practice phase is the most important phase of the microteaching programme. In this phase the teacher trainee chooses a specific skill, prepares a micro-lesson plan and teaches a small group of students for a duration of 5 - 6 students. We follow few steps to get training in the skill. These steps are called microteaching cycle.



Steps in microteaching cycle

1. **Selecting a skill** : This involves selection of a teaching skill practiced, knowing its components clearly.

2. **Observing model lesson** : Teacher educator will present a model microteaching lesson plan of the selected skill. Trainee observe it carefully
3. **Preparing Lesson Plan** : Teacher trainee has to prepare a micro-lesson plan for 5 - 6 minutes for the selected teaching skill. In lesson plan all the components of the skill should be presented in an appropriate place.
4. **Teaching** : The trainee teaches the lesson for 5-6 minutes. There will be 5-6 students or peers with teacher educator or one or two peers as supervisor. If possible record the teaching methods of the trainee by video or audio tape.
5. **Feed back**: The supervisor analyses the performance and discusses it with the teacher trainee. The feedback should focus on components of the teaching skill and draw attention to modification for mastering the skill.
6. **Replanning**: In the light of the feedback received from the supervisor or peer observers, the teacher trainee replans her micro-lesson plan by writing another lesson plan or modifying the existing one on the basis of feedback.
7. **Reteaching**: The teacher-trainee reteaches the revised lesson plan to another group of students. The supervisor checks to see whether there is any improvement in skill attainment. This process repeats till the teacher trainee acquires the required level of competency.

2.1.5 Uses of microteaching

Microteaching techniques increases the effectiveness of the teacher training programmes in the following ways

1. Microteaching helps in reducing the complexities of the normal classroom teaching. This helps the teacher trainees gain more confidence in classroom teaching.
2. Microteaching gives the knowledge of various teaching skills.
3. It is useful for developing teaching efficiency.
4. Microteaching prepares trainee to effective and better teacher.
5. Microteaching creates among the teacher trainees an awareness of the various skills of which teaching is composed of.
6. Each trainee makes progress in developing teaching skills at his own speed depending his ability.
7. Using video or Audio tape enable the trainee to analyse his own teaching performance.
8. Feedback enable the teacher-trainee to consciously concentrate on specific behaviour modification.\

2.1.6 Micro Teaching Vs Macro Teaching

Sl.No.	Micro Teaching	Macro Teaching
1.	It is a teacher training technique.	It is a method of classroom teaching
2.	Class size is 5 to 10 students	Class size is more than 40
3.	Duration of the class 6-10 minutes	Duration of the class 40 to 45 minutes
4.	One or two teaching Skills used	Maximum Teaching skills are used.

Sl.No.	Micro Teaching	Macro Teaching
5.	Importance is given to skill	Importance is given to content
6.	It puts the teacher under the microscope	No such arrangement for macro teaching
7.	All the faults of the teacher are observed	No such arrangement for macro teaching
8.	Feed back is given	No feed back is given
9.	Opportunity to reteach the lesson	No such arrangement
10.	No importance is given to students needs	Importance is given to students needs
11.	No classroom activities for students	Classroom activities will be given.
12.	It tends to reduce the creativity of the teachers	It increases the creativity of the teachers
13.	Micro teaching can be successful in a controlled environment only	Macro teaching can be successful in any environment
14.	Teaching is relatively simple	Teaching becomes complex
15.	Micro teaching is done with all sensory resources	Macro teaching relies on auditory learning and recall.

2.2 Identification of teaching skills:

A teaching skill is a set of activities or actions of the teacher which are repeated a number of times during the course of teaching.

Characteristics Of Teaching Skills

- a. Teaching skills are the specific instructional activities.
- b. Teaching skills are verbal and non-verbal in nature.
- c. A teaching skill includes activities and actions.
- d. Teaching skills involve the teacher's activities which are performed in classroom.
- e. Teaching skills develop the communicative competency.
- f. Teaching skills help effective presentation of content in classroom.

List of Teaching skills given by Allen and Ryan

- 1. Stimulus variation
- 2. Set induction
- 3. Closure
- 4. Teacher silence and non-verbal cues
- 5. Reinforcing pupil participation
- 6. Fluency in questioning
- 7. Probing questioning
- 8. Use of higher order questions
- 9. Divergent questions
- 10. Recognising and attending behaviour
- 11. Illustrating and use of examples
- 12. Lecturing

13. Planned repetition

14. Completeness of communication

List of Teaching skills given by B.K. Passi

1. Writing instructional objectives
2. Introducing a lesson
3. Fluency in questioning
4. Probing questioning
5. Explaining
6. Illustrating with examples
7. Stimulus variation
8. Silence and non-verbal cues
9. Reinforcement
10. Increasing pupil participation
11. Using blackboard
12. Achieving closure
13. Recognizing attending behaviour

Types of Teaching Skills selected

Large number of teaching skills have identified in bringing effective teaching. Allen and Ryan listed fourteen teaching skills and in 1983 Menon gave a list of seventy four teaching skills. A few important skills listed below will be discussed here.

1. Skill of Introducing the lesson / Skill of set induction
2. Skill of explaining
3. Skill of questioning
4. Skill of probing questions

5. Skill of stimulus variation
6. Skill of reinforcement
7. Skill of Non-verbal clues
8. Skill of closure and

2.2.1. Skill of Introducing the Lesson / Skill of set induction

In teaching, the way of introducing a lesson is more important. The proverb "Good beginning is half of the work completed." is also fit for teaching. The teacher, before introducing a lesson, has to motivate the students and establish cognitive and affective rapport with the students. The following are the components of skill of introducing the lesson.

1. Testing previous knowledge
 - a. Preliminary attention gaining
 - b. Use of previous knowledge
 - c. Use of appropriate device
 - d. Link with a new topic
2. Attention Gaining
 - a. Use of voice to focus attention
 - b. Use of audio visual aids
 - c. Use of gesture / eye contact
 - d. Introducing something unusual
3. Motivation stimulation
 - a. To arouse curiosity
 - b. Method of introducing a lesson
 - c. Able to involve students

1. Testing previous knowledge**1a. Preliminary attention gaining**

Tell the significance of the topic. Give importance of new knowledge in our life situations.

1b. Use of previous knowledge

Before giving new knowledge, teacher has to test the previous knowledge which is to be used in the new topic.

1c. Use of appropriate device

By way of asking few questions or asking students to write two or three sentences about the new topic.

1d. Link with a new topic

Integrate the old knowledge with the new knowledge. Use the knowledge obtain from Text book, Library book etc., to teach the new topic.

2. Attention Gaining**2a. Use of voice to focus attention**

Use clear and bold voice to focus attention.

2b. Use of audio visual aids

Use appropriate teaching aids required to teach the topic to arouse pupil's attention. Charts, slides and power point presentation may be used.

2c. Use of gestures / eye contact

Gestures are indication of conveying messages by body movements. Eye contact is a signal for lack of interest in a person. Much of eye contact shows that it is a threatening attitude.

2d. Introducing something unusual

Relevant to the new lesson / topic, show a chart, photograph or a instrument which is not seen by student,

3. Motivation stimulation**3a. To arouse curiosity**

To bring in interest on the topic the teacher must refresh the students properly. He / She should enact to the situation. Must quote a lot of day to day happenings of the new knowledge and give a lot of examples. Self involvement is needed to bring in interest among students. He must become a part of lesson, only then he can arouse curiosity in the students.

3b. Method of introducing a lesson

Special care should be given to introduce a lesson. Teacher has to select any method / technique to introduce a lesson from the methods given below according to the nature of the lesson.

1. Telling relevant statements to teach the new topic.
2. Introduce the lesson after obtaining answers from the students.
3. By way of anecdotes, Explaining or commenting, introduce a lesson.
4. By story telling technique, introduce a lesson.
5. By the method of dramatisation, introduce a lesson.
6. Using teaching aids, we introduce a lesson.
7. By the method of demonstration, introduce a lesson.

3c. Able to involve students

While introducing a lesson teacher can ask students to collect or prepare materials / Aids well in advance.

2.2.2. Skill of Explaining

For an effective learning, explanation in teaching should be done effectively. Explaining bridges the gap in understanding the new knowledge by relating it to the past knowledge. Thus explaining depends upon the past knowledge, the type of new knowledge and the relationship among them. Explanation can be more effective by using simple and clear language for clarity. The following are components of the skill of explanation.

1. Beginning statement
2. Explaining links
3. Concluding statement
4. Questions to test pupil's understanding
5. Irrelevant statements
6. Lacking in continuity
7. Vague words and phrases

1. **Beginning statement:**

While explaining the beginning statement must include central concept of the content. Statements such as 'Today we are going to derive a formula or prove a theorem' etc. Beginning statement must gain attention of the students.

2. **Explaining links:**

While explaining the subject matter use suitable link words between the concepts. examples: therefore, that is why, since, because, hence, in order to etc.

3. Concluding statement:

Concluding statement should summarise what the teacher explained. Students can summarise what they learn if concluding statements are meaningful.

4. Questions to test pupil's understanding

Questions asked here are from the content explained. Teacher check the understanding of students by questioning. These questions will also help to analyse the performance of the explanation given by the teacher.

5. Irrelevant Statement

Teachers should avoid irrelevant statements while explaining the topic.

6. Lacking in continuity

Knowledge given in classroom are information, facts, concepts etc. It should be given in sequence and in continuity. Then only the students clearly understands the concepts, facts and information.

7. Vague words and phrases

The teacher should not use vague words and phrases which is not understandable by students. The vocabulary used should be clear and according to the level of students.

2.2.3 Skill of questioning

Successful teaching technique is highly depend on questioning. It is a skill that can be used in teaching any subject and any standard. In a person different types of questions, depending upon situation and purpose to be achieved are used. Questions are used to help students recall certain facts, exercise their reasoning ability, and use

their power of recognition and discrimination in enhancing their learning. Effective questioning encourages them to reflect over things and ideas and leads them to participate in discussion.

Purpose of questioning

- Questioning is a logical procedure of problem solving
- Questioning is useful for presenting the tentative nature of knowledge.
- The teacher encourages the learners to seek more than one answer for a question
- The teacher uses questioning to achieve learning objectives
- The teacher uses questioning to assess students understanding.

Thus questioning promotes involvement, enhance learning and motivate students.

Let us examine why questioning is an essential skill for a teacher.

- It helps students test their knowledge
- It explicit desired information from students
- It stimulates thinking
- It develops subject matter
- It enable students to analyse the problem
- It enable students to apply their knowledge to a specific new situation.
- It helps students evaluate for themselves their understanding of the subject matter.

- It motivate students to participate in the teaching-learning process.

Questioning stimulates thinking. The level of students thinking depends upon the level of questions asked by the teacher. The level of question is also determined by the level of the responses given by the students. In order to elicit responses at different levels from the students, we can frame questions at different levels.

Questioning is one of the most important skill in teaching. It has to be developed by any teacher. While teacher teaches a lesson he has to ask several questions related to the content. These type of questions are called teaching questions. After completing a lesson, questions will be asked to know whether the students understood the content of teaching. These questions are called review questions. Teachers should be expert in asking questions. Good questions serve many purposes. Questions stimulate thinking. Questions help recalling facts, Concepts and principles. Questions make active involvement in learning. Questions arouse the curiosity of the students

Generally these are three levels of question. They are 1. Low order questions 2. middle level questions 3. higher order questions.

Low order questions.

These questions require the students to recall knowledge that they have learnt. The questions of this level do not ensure understanding of the content or analyzing information, but requires memorization.

Examples : Give the properties of an algorithm.

State any two differences between definite and indefinite iterations.

How will you know a hidden file name.

Intermediate level of questions.

Explaining the meaning of a concept or statement, compare or differentiate among two things etc are intermediate level of questions.

1. Write c program to find the area of a triangle.
2. Write a c program to count the number of vowels present in your name.
3. Explain the function of **Man**.

Higher order questions

These type of questions requires the students to use higher levels of intellectual abilities such as comprehension, application, analysis, synthesis and evaluation.

Example:

1. Write a programme to convert a binary number into its equivalent decimal.
2. Distinguish between CD-RW and CR-R?

2.2.4 Skill of probing

The skill of probing questions involves going deal into pupil's response by asking a series of subsequent questions in a topic. It helps the pupils to give a correct and complete answer. This skill requires the teacher to put a series of questions about the answer given to the first question. It is a major micro-teaching skill.

When students are unable to give the answer or give incomplete answers, teachers use probing questions.

Purpose of probing

1. To motivate the students for correct answer

2. Ask many questions related to the main subject
3. Taking various supportive questions
4. Linking previous knowledge with the new knowledge
5. Creating interest and curiosity among students.
6. Developing the thinking, creativity and analytical ability

Components of probing skill of questioning

There are five main components of skill of probing

1. Prompting
2. Seeking further information
3. Re-Focusing
4. Re-Direction
5. Critical awareness

1. Prompting

Prompting mostly involves verbal questions, visual questions and some visual guidance. In order to help the student to know the correct answer, the teacher gives some indication.

2. Seeking further information

Most of the time student's don't give the proper and accurate answer. Then the teacher asks the same question to other students in order to get the correct answer. The teacher asks different aspects of the question in order to seek further information. This also may check the students' knowledge.

3. Re-Focusing

If the teacher is not satisfied with the answer of the student then the teacher puts different situations in front of

the students which helps them in the transfer of knowledge of the students.

4. Re-Direction

The teacher tries to know the multiple answer to the question in order to make students think again and again. This will increase the thinking ability of the students, and this may increase the participation of the students in the class too.

5. Critical awareness

If the teacher asks why? how? type of questions then it is called Critical Awareness. This critical awareness among students increase the rational thinking and awareness.

2.2.5. Skill of stimulus variation

While teaching teacher must draw attention of the students. For this, teacher uses some gestures, body movements, makes certain verbal statements, modulating the voice etc. All these behaviours are related to stimulus variation. Components of this skill are :

1. Movements
2. Gestures
3. Change in speech pattern
4. Change in inter action style
5. Secure attention
6. Pausing
7. Oral - Visual switching

1. Movements

Teacher should not stand in one place while teaching. He has to move from one place to another to write on the blackboard, to go near and appreciate a student, to move

towards a model etc. Hence, in order to gain students attention a teacher has to move purposeful movements.

2. Gestures

Use of appropriate gestures according to the content to emphasis importance, to explain emotions, make students to understand the message. Hand movement to describe shape, size, Body movements and facial expression enhance the value of message.

3. Change in speech pattern.

If a teacher explains the content in the same modulation, then it is boring. Hence, in teaching there should be voice modulation to pay attention to a particular point. Modulating the voice on situations, change in pitch, speed of voice will force his student to pay attention.

4. Change in interaction style

Teacher has to plan, how students also involve in teaching. Asking questions from the students, asking student to do a sum on the blackboard, students are encouraged to put few questions and teacher will answer it, to participate students in a seminar or discussion and teacher will tell its result etc. are all called change in interaction. This can be mention in a simple form as

Teacher - Class

Teacher - Student

Student - Student

5. Secure Attention.

While teaching the teacher may use certain statements like, this is important, look at this diagram, listen carefully etc to draw attention. While writing on the blackboard teacher draw attention by using colour chalk for important message or underlined the important points.

6. Pausing

Pausing means introducing silence during teaching. From this act, we can draw attention of the students. While teaching continuously or at the time of asking questions teacher has to plan pausing and create attention.

7. Oral - Visual switching

While the teacher orally explains, student's hearing. When the teacher exhibits aid or writes on blackboard his visual sense is used. Such frequent changes from oral to visual or from visual to oral help to secure pupil's attention.

2.2.6 Skill of Reinforcement / Skill of Appreciation

All these behaviours are related to stimulus variation. The skill of reinforcement refers to the effective use of reinforcers. While teaching, a teacher asks questions and students replies it with correct answer. If the teacher simply goes to another question, then students will not happy. If the teacher reinforce a correct answer with a smile or praise the student, then they will be happy. This is called reinforcement. Teacher has to practice using positive reinforcers and avoid the use of negative reinforcers in the classroom. By Reinforcement, students participation is more in classroom activities. The different components of this skill are

1. Positive verbal reinforcement
2. Positive non verbal reinforcement
3. Negative verbal statement
4. Negative non verbal reinforcement
5. Wrong use of reinforcement
6. Inappropriate use of reinforcement

1. Positive verbal reinforcement

When the students give correct response for a question the teacher has to give positive verbal reinforcement such as Good, Very good, Well, Excellent, Continue etc

2. Positive non verbal reinforcement

When a student give correct response for a question the teacher may give non-verbal reinforcement expressing happiness or with the help of body language. Teachers should use more positive reinforcers so that the students participation is maximum in teaching activities.

3. Negative verbal statement

Teacher should avoid using these type of negative verbal reinforcers namely, I do not like you, stop it, nonsense, you won't improve, you will fail in the examination etc.

4. Negative non verbal reinforcement

Teacher should not use these type of Non-verbal negative reinforcers such as facial expression like, frowning, sadness, moving away from responding student, looking at angrily etc.

5. Wrong use of reinforcement

When a student gives a wrong answer, teachers should avoid using words like idiot, stop, enough or showing terror face etc. which will hurt student and his enthusiasm will decrease in the subject.

6. Inappropriate use of reinforcement

Teacher should not use reinforcers in a delayed manner or using it always. Using the above type of reinforcement

affects the learning and decrease the motivation of the students.

3.2.7. Skill of Non-Verbal cues

Non-verbal cues has been defined as communication without words. Although verbal output can be turned off, non verbal can't. Teachers use Non verbal cues because

1. Words have limitations. There are numerous areas where non verbal communication is more effective than verbal.
2. Non verbal signs are powerful. Non verbal cues primarily express inner feelings.
3. Non verbal messages are likely to be more genuine.
4. Non verbal signals can express feelings, thoughts etc.
5. Complexity of the verbal message can be given through simple non verbal signals.

Components of the skill

1. Body language
2. Proximity
3. Touch
4. Eye contact
5. Chronemics
6. Smell
7. Vocalics

1. **Body language :** Body movement have specific verbal transactions. To explain the subject matter or to appreciate the students we use body language. To describe shape, size body movements are helpful.

2. **Proximity:** When a teacher wants to convey message to a student or to control a problematic behaviour student he has to go near him and convey non verbally.
3. **Touch :** Physical contact with others is the most basic form of communication and certain situations often communicates that there is a problem.
4. **Eye contact :** It is a signal for lack of interest in a person. Too much eye contact shows that it is a threatening attitude.
5. **Chronemics:** With the help of chronemics we can draw attention of the students by means of waiting and pausing. While teaching teacher has to plan waiting and pausing. By increasing the waiting time, you buy for yourself an opportunity to hear and to think. Increase your waiting time upto 5 seconds or longer for slow and shy students and increase in waiting time help students to respondent higher cognitive skills.
6. **Smell :** We can make identify things using smell. This can be used largely in lower standards.
7. **Vocalics :**

Postures : Postures can be used to determine participants degree of attention or involvement. Posture is understand through such indicators as direction of learn body orientation, arm position and body openness.

Adornment : Teacher should wear decent dress. If the school introduce uniform to teachers, teachers should strictly follow them. Lady teachers should not come with more jewels and shining dresses. Clothing should reflect our culture. Hair style should not be funny or attractive.

Locomotion : Teacher should not stand or sit in one place during teaching. He has to move one place to another place or go near to a student to convey non verbal messages.

3.2.8. Skill of closure

After teaching a subject matter in a better way, it is better to close it in the same way. Then only the students feel pleasure and satisfaction. At the end of the lesson, it is necessary to make a quick review. What has been taught, applying new knowledge in various situations, associating new knowledge with old knowledge and repeating important points. Various components of this skill are:

1. Consolidation of major points.
2. Application of present knowledge.
3. Linking post knowledge with present knowledge.
4. Linking present knowledge with future learning.

1. Consolidation of major points

While teaching teacher has explained so many facts to explain the subject matter. At the end he has to summarise it in a meaningful way. Then only students identify what they have learned.

2. Application of present knowledge

After teaching new knowledge, the teacher has to explain how and where it is used with examples. Then only the students knows the importance of new knowledge and its application. For this the teacher may use problem solving method.

3. Linking past knowledge with present knowledge

After explaining the new knowledge, teacher should link it with the past knowledge. Then only the students get a clear view about what they learned and feel that their knowledge has developed.

4. Linking present knowledge with future learning

The new knowledge learned by the students should be linked to the knowledge what they will going to study in future or in continuation to this knowledge. From this they will ready to learn more about this knowledge.

2.2.10 Advantages or merits of Micro teaching

- It offers opportunity to practice a real lesson without the complexities of a normal classroom.
- It offers several sources of immediate feedback.
- One skill is learnt at a time which is observable, controllable and practicable.
- It helps in the development of self-confidence.
- It helps the trainees to develop the skill of blackboard writing.
- It is helpful to improve his/her own teaching.
- It makes trainees familiar with various teaching styles, eg. Black-board writing, use of aids, asking questions etc.
- Evaluation procedure of a micro-lesson is more precise, objective and acceptable.
- It saves time and energy of the student-teacher as well as the teacher - educator.

- It focusses attention on the modification and improvement of teacher behaviours in the teaching learning process.

2.3. Model Micro Lessons

Model Micro Lesson – 1

Name of the Student Teacher : xxxxx

Skill : Skill of Explaining

Standard : 11th

Subject : Computer Science

Unit : Computer Communications

Topic : Types of Network

Duration : 5 Minutes

Date : xxxxx

Components of skill

1. Beginning statement (1)
2. Explaining links (2)
3. Concluding statement (3)
4. Questions to test pupil's understanding (4)
5. Irrelevant Statements (5)
6. Lacking in continuity (6)
7. Vague words and phrases (7)

Episode

Teacher : Good Morning Students

Student : Good Morning Teacher

Teacher : Today we are going to see types of network.
(1)

The following are the general types of network used today.

1. Local Areas Network (LAN)
2. Metropolitan Area Network (MAN)
3. Wide Area Network (WAN)

How many types of network are used? (4)

Student : 3 types of network.

Teacher : What are they? (4)

Student : LAN, MAN and WAN

Teacher : A network connecting systems and devices inside a singly building or buildings close to each other is called Local Area Network (LAN). Generally LAN does not use the telephone network. (2)

LAN stands for?

Student : Local Area Network

Teacher : LAN connected either by wire or wireless.

Wired connection may be using twisted pairs, coaxial cables or fiber optic cables. In a wireless LAN, connections may be using infrared or radio waves. Wireless networks are useful when computers are portable. However, wireless network communicates slowly than a wired network. (2)

Give some example of wire? (4)

Student : Twisted pairs, coaxial cables and fiber optic cables.

Teacher : The number of computers in the network is between two to several hundreds. LAN is

generally used to share hardware, software and data. A computer sharing software package and hard disk is called a file server or network server.

What is a file server? (4)

Student : A computer sharing software package and hard disk is called a file server.

Teacher : A network that spans geographical areas covering a Metropolitan city is called Metropolitan Area Network (MAN).

MAN stands for? (4)

Student : Metropolitan Area Network.

Teacher : A WAN is typically two or more LANs connected together across a wide geographical area. The individual LANs separated by large distances may be connected by dedicated links, fiber-optic cables or satellite links.

What is meant by WAN? (4)

Student : WAN means two or more LANs connected together across a wide geographical area.

Teacher : Today we have seen the types of networks (LAN, MAN, MAN). Tomorrow we will see Network Topology.

Thank you students

Student : Thank you teacher

Model Micro Lesson – 2

Name of the Student Teacher : xxxxx

Skill : Skill of Stimulus Variation

Standard : 11th

Subject : Computer Science

Unit : Operating System

Topic : Process Management

Duration : 5 Minutes

Date : xxxxx

Components of skill

1. Movement (1)
2. Gestures (2)
3. Change in speech pattern (3)
4. Change in interaction (4)
5. Secure attention (5)
6. Pausing (6)
7. Oral visual switching (7)

Episode

Teacher : Good Morning Students

Student : Good Morning Teacher

Teacher : Today we are going to see about the process of management.

The teacher moves to write the topic on the blackboard. (1)

Process management undertakes the allocation of processors to one program. The operating system controls the jobs submitted to the system (CPU). CPU stands for? (4)

Student : Central Processing Unit.

Teacher : Write the answer on the blackboard. (7)

Teacher : Several algorithms are used to allocate the job to the processor.

Teacher changes the voice and stop. (6)

Algorithm is a step-by-step method to solve a given problem.

What is algorithm?

Student : Gives correct answer.

Teacher : Listen carefully. (5) There are four types of algorithm.

1. FIFO
2. SJF
3. Round Robin
4. Based on Priority (1) (2) (3)

How many types of algorithm in process management?

Student : 4 types

Teacher : Look at the blackboard. (5) FIFO stands for First In First Out. This algorithm is based on queuing. Suppose you are standing in a queue

to get your notebook corrected from your teacher. The student who stands first in the queue gets his/her notebook corrected and so on. This is the basic methodology of the FIFO algorithm. (1) (2)

FIFO stands for? (4)

Student : First In First Out. (7)

Teacher : Now, let us deal with this FIFO a little more technically. (5) the process that enters the queue first is executed first by the CPU, then the next and then next and so on. The processes are executed in the order in which they enter the queue. (1) (2) & (6)

What is the process of FIFO? (4)

Student : Gives correct answer.

Teacher : Second one is SJF (Shortest Job First). This algorithm is based on the size of the job.

Take two jobs A and B.

A = 5 kilo bytes

B = 8 kilo bytes

Kilo literally means 1000 but here kilo means 1024. A byte consists of eight bits.

Teacher : Today we have seen only two types of algorithm in process management.

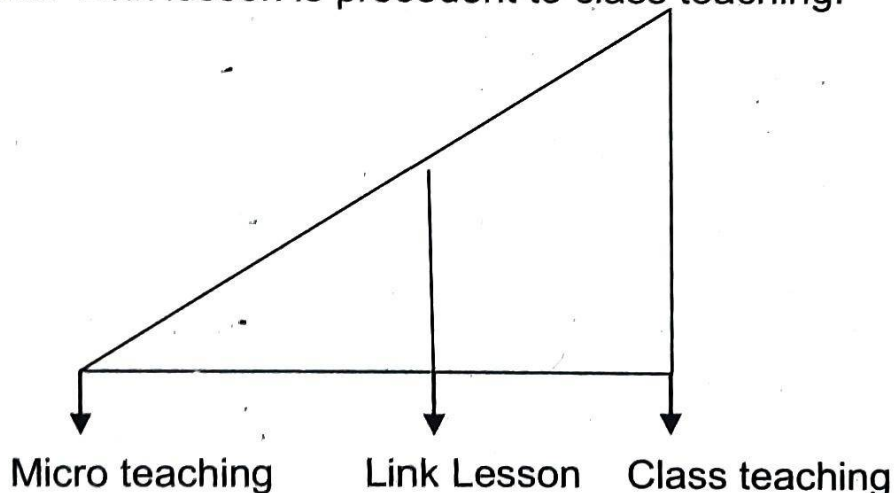
Tomorrow we will see the remaining types of algorithm.

Thank you students

Student : Thank you teacher

2.4 Link Lesson

In micro teaching each teaching skill is taken separately and training is given in that skill. All these skills are combined in class teaching. In micro teaching importance is given to teaching skill then the content. In Link Lesson a small content and three or four teaching skills are taken. Importance is given for both content and teaching skills in link lesson. Here teacher trainee is teaching 10 - 15 students in 10 - 15 minutes. In this way if we give few teaching classes to the teacher-trainee to take link lessons and then if we send to class teaching, then it will be more useful. Link lesson is precedent to class teaching.



Micro teaching → Link Lesson → Class teaching

Understanding major Steps in teaching a Link Lesson

While preparing a Link lesson plan, the following steps are followed :

Designing a Link Lesson

The following steps are used in designing a Link Lesson plan.

1. Selecting a content / topic.
2. Setting Lesson objectives.
3. Resource materials needed.
4. Content outline
5. Teaching skills used
6. Instructional Procedures

1. Selecting a Content / topic

The first thing to consider obviously is what you want to teach. This should be based a unit / topic of your syllabus. You also need to be aware of what standard you are developing a topic plan and also the time given for your teaching.

2. Setting Lesson objectives

To make sure your topic plan will teach exactly what you want it to do, you need to develop clear and specific objectives. Note that objectives are the learning outcomes of the lesson. You can certainly have more than one objective for a topic plan.

3. Resource materials needed

Find out what resource materials you are giving to use to explain the topic plan and prepare it well in advance. Be sure the teacher will have materials everything to teach the topic. Power point presentation, Flash Cards, charts or materials are very helpful in teaching your topic.

4. Content outline

Write content key points to be presented.

5. Teaching Skills used

c

Write the name of the teaching skills used. While teaching a link lesson.

6. Instructional Procedures

- a. Motivation
- b. Presentation
- c. Interaction
- d. Reflection
- e. Summing up

a. Motivation

Students are encourage and motivate to learn new knowledge. The mind of the child should be prepared to receive new knowledge. Preparation may involve.

1. Testing previous knowledge to learn new knowledge.
2. Integrating the previous knowledge with the new knowledge.
3. Arrest attention of the students using new techniques.
4. Announcing the aim of the lesson.

b. Presentation

Here teaching takes important place. Students learn new knowledge. For an effective learning outcome, teacher should ensure active student participation with the use of audio-visual aids. Teacher should stimulate thinking by asking number of questions while teaching.

c. Interaction

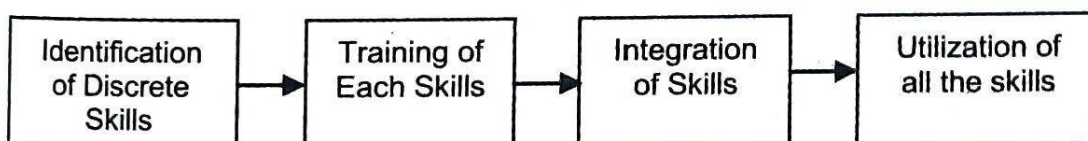
Interaction means the student teacher activities while teaching is going on in the class. Interaction leads learning. All interaction will take place to attain the objectives.

d. Reflection

A process of self-observation and self-evaluation is called reflection. By collecting information about what goes on in our classroom and by analysing and evaluating this information is called reflection. We identify and explore our success and failures. This may then lead to changes and improvements in teaching. Reflection can be given by self, peer people or students. Reflection can also be seen with the help of audio and video recording of class teaching.

e. Summing up

After teaching a subject matter in a better way, it is better to close it in the same way. Then only the students feel pleasure and satisfaction. At the end of the lesson, it is necessary to make a quick review what has been taught, and repeating important points.

Practising a Link Lesson**Concept of Link teaching****Link Lesson plan**

A Link lesson is a short lesson that provides instruction in one or more skills or concept that students will relate to a larger lesson plan that follow. Link lessons can be used to

teach particular skills, create interest in a topic and generate questions or introduce strategies.

The art of teaching, writing, the Link lesson allows a teacher to convey a content or strategy to students that they will use often. It allows students to gain valuable relevant skills on a regular basis without spending too much of time to teach the same skills. The Link lesson should be short and its duration will be 15 to 20 minutes. Here the teacher introduce the topic, demonstrate skills or concept; guide student practice; discuss the topic; give more examples; and task about what was taught. Link lessons can be taught individually, in a small group or to a whole class.

Link lesson plan format

Name of the Teacher Trainee :

Date :

Standard and section :

Subject : Computer Science

Unit :

Topic :

Instructional objectives : 1.

2.

3.

Instructional Resources Required

1.

2.

Content outline

Teaching Skills used**Instructional procedures****Link Lesson - Lesson Plan**

In the beginning section of this unit we have learnt various skills and their components. This must have significantly contributed to the qualitative improvement of teaching. Now we can integrate all those skills and use them in our teaching. It helps trainee to transfer effectively all the skills.

As episode learnt in the micro teaching on integration of teaching skills is given below for your guidance. Student teachers are asked to study it carefully and plan their practice of integration of teaching skills.

Skills to be integrated

Following skills are integrated and an episode is prepared.

1. Skill of Introducing the lesson
2. Skill of explaining
3. Skill of questioning
4. Skill of probing questions
5. Skill of stimulus variation
6. Skill of Non-verbal cues
7. Skill of Reinforcement
8. Skill of closure

An episode linking all the above skills is given here under.

Model Link Lesson Plan

Name of the Teacher Trainee : xxxxx
Date : xxxxx
Standard and Section : XI C
Subject : Computer Science
Unit : Computer Organization
Topic : Memory Unit

Instructional objectives

1. Acquires knowledge about memory unit.
2. Understands the types of memory.
3. Explains the process of memory.
4. Realize the benefits of memory.
5. Develops skill of how to work with the memory of the computer.

Instructional Resources Required

1. Chart (types of memory)
2. Models (chip)

Teaching skills used

1. Skill of introducing the lesson
2. Skill of explaining
3. Skill of questioning
4. Skill of stimulus variation
5. Skill of non-verbal cues
6. Skill of reinforcement
7. Skill of closure

Content outline

Meaning of memory unit – types of memory – process of memory – benefits of memory – application – recapitulation – follow up activities

Content/ Concept	Specification of behavioural objectives	Learning Experience (Teacher/Learner activities)	Evaluation
1. Motivation	Recall and Motivates	The teacher explains some storage devices. Such as Tape recorders, Mobile phones, CD and computers. Teacher shows the model of storage devices and explains its benefits. Discusses about human memory.	What are the storage devices? What are the benefits of the storage devices? Can you tell me the uses of the human memory?
2. Presentation Meaning of memory unit	Acquires knowledge	Memory units are the storage areas in a computer. The term memory usually refers to the main memory of the computer, whereas, the word storage is used for the memory that exists on disks, CDs, floppies or tapes. The main memory is usually	What is memory unit? What is use of the main memory?

Content/ Concept	Specificati on of behaviour al objectives	Learning Experience (Teacher/Learner activities)	Evaluation
Type of Memory	Understands	called a physical memory which refers to the chip (Integrated Circuit) capable of holding data and instruction. The teacher explains the types of memory with the help of chart. 1. Random Access Memory (RAM) 2. Read only Memory (ROM) 3. Programmable Read only Memory (PROM) 4. Erasable Programmable Read only Memory (EPROM) 5. Electrically Erasable Programmable Read only Memory (EEPROM)	How many types of memory? What are they?

Content/ Concept	Specification of behavioural objectives	Learning Experience (Teacher/Learner activities)	Evaluation
Process of Memory	Explains and Realizes	Random Access Memory – RAM RAM is the most common type of memory found in the modern computers. This is really the main store and is the place where the program gets stored. When the CPU runs a program, it fetches the program instructions from the RAM and carries them out. If the CPU needs to store the results of the calculations it can store them in RAM. When we switch off a computer, whatever is stored in the RAM gets erased. It is volatile form of memory.	What is the process of RAM?

Content/ Concept	Specification of behavioural objectives	Learning Experience (Teacher/Learner activities)	Evaluation
		Read only Memory – ROM In ROM, the information is burnt into the ROM chip at manufacturing time. Once data has been written into a ROM chip, it cannot be erased but you can read it. When we switch off the computer, the contents of the ROM are not erased but remain stored permanently. ROM is a non-volatile memory. ROM stores critical programs such as the program that boots the computer.	What is use of ROM?
3. Interaction	Give examples	Different types of storage devices and its benefits The teacher shows the mobile phone memory	How much storage capacity of your mobile phone's memory

Content/ Concept	Specificati on of behaviour al objectives	Learning Experience (Teacher/Learner activities)	Evaluation
		card and interacts with the students about its process.	card?
4. Reflection	Recalls and Recognizes	Students give the correct responses for all the questions.	What do you mean by memory unit? What are the types of memory? ROM Stands for? RAM Stands for?
5. Summing up	Recalls and realizes	At the end, the teacher recalls all the important points and again explained. We will see the remaining types of memory in next class.	

Follow up activities:

1. Collect the picture of types of memory?
2. Write down the benefits of storage devices?
3. Study well the functions of memory unit?

2.5 Link Lesson Episode

Episode

Skill	:	Integration of skills
Class	:	XI
Subject	:	Computer Science
Unit	:	An introduction to windows XP
Topic	:	What is Windows XP

Teacher / Student	Teaching – Learning Activities	Skill Components
Motivation Teacher	Good Morning Students.	
Students	Good Morning Sir.	
Teacher	OS stands for?	Skill of introducing
Students	Operating System	
Teacher	Give some examples for OS?	Skill of introducing
Students	MS-DOS, UNIX, LINUX	
Teacher	Very good. Another one?	Skill of reinforcement, Skill of questioning
Students	Windows XP	
Presentation Teacher	Today we are going to see about what is Windows XP. Windows XP Professional is a user-friendly operating system designed for	Skill of explaining, Skill of stimulus variation, Skill of fluency in communication

Teacher / Student	Teaching – Learning Activities	Skill Components
	popular use. The most important advantage of using windows is its GUI. GUI stands of Graphical User Interface.	
Interaction Teacher	GUI stands for?	Skill of questioning
Students	Graphical User Interface.	
Teacher	Good. CUI stands for?	Skill of reinforcement, Skill of questions
Students	Character User Interface	
Teacher	GUI is said that the right side brain is good in processing the pictures and is the seat of creative thinking and intuitive ideas whereas the left side brain is good at logical thinking. (use chart)	Skill of explaining, Skill of stimulus variation Skill of non-verbal cues
Teacher	It is believed, before the introduction of GUI, users of OS, mainly used their left side brain, keeping	Skill of fluency in communication

Teacher / Student	Teaching – Learning Activities	Skill Components
	their right side of the brain idle. It is felt, windows effectively uses the left and right side of the brain.	
Teacher	Many of other operating system use command line interface. Interface is a connection with the machine. For example T.V. remote (show remote) Another one example?	Skill of explaining Skill of non-verbal cues Skill of questioning
Student	Telephone and all other remotes	
Teacher	Smiling.	Skill of reinforcement
	In this kind of interface, you have to remember cryptic commands and type them without mistakes. To make things worse some operating systems are case-sensitive also. What is meant by Case-sensitive?	Skill of explaining, Skill of stimulus variation Skill of questioning

Teacher / Student	Teaching – Learning Activities	Skill Components
Student	It means small letters and capital letters are not same.	
Teacher	Excellent. A simple spelling mistake or missed space will result in an error. Windows displays all the information on the screen and all you have to do is to point and select using the mouse, with its GUI. A picture is worth a thousand of words, as they say.	Skill of reinforcement Skill of fluency in communication
Summing up Teacher	Windows XP Professional combines all the positive aspects of its Microsoft predecessors. This satisfies all the users who want to prevent frequent crashing of software and want to use easy techniques.	Skill of stimulus variation
	Tomorrow we will see evolution of the operating system.	Skill of closure

Difference between microteaching, Link lesson and class teaching

Characteristics	Micro teaching	Link lesson	Class teaching
Time	5 - 6 minutes	15 - 20 minutes	40 - 45 minutes
No. of students	5 -6	10 -15	40 - 50
Content	Very Small content	Small content	One Lesson or part of it
Teaching skills	Only one skill	Three or four skill	Various skills

REVIEW QUESTIONS

1. Explain in detail the Micro teaching cycle.
2. Explain in detail the skill of set induction with its components and give an episode.
3. Explain the Skill of Explaining with its components.
4. Write the difference between Micro teaching and Macro teaching.
5. Write the major steps in teaching a Link Lesson.
6. Write a Link lesson episode for XI Standard Computer Science.